

Exercise 1: Marchenko–Pastur Law

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Problem 1 (Stieltjes transform):Let μ be a probability measure on $\mathbb{R}$ and define its Stieltjes transform by

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t-z} \mu(dt), \quad z \in \mathbb{C} \setminus \text{supp}(\mu).$$

Compute $S_\mu(z)$ explicitly in the following two cases:

1. $\mu = \delta_a$, the point mass at $a \in \mathbb{R}$.
2. μ is the uniform distribution on $[0, 1]$.

Problem 2 (Stieltjes transform at infinity):Let μ be a probability measure on $\mathbb{R}$. Define its Stieltjes transform by

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t-z} \mu(dt), \quad z \in \mathbb{C} \setminus \text{supp}(\mu).$$

Show that, as $|z| \rightarrow \infty$,

$$S_\mu(z) = -\frac{1}{z} + O\left(\frac{1}{z^2}\right).$$

Hint. Write

$$\frac{1}{t-z} = -\frac{1}{z} \frac{1}{1 - \frac{t}{z}} = -\frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{t}{z}\right)^k,$$

and justify term-by-term integration for $|t| < |z|$.**Problem 3 (Simulation of Eigenvalues):**

Write a short program to generate

$$\hat{\Sigma}_n = \frac{1}{n} \mathbf{X} \mathbf{X}^\top$$

with $X \in \mathbb{R}^{p \times n}$ having i.i.d. $\mathcal{N}(0, 1)$ entries, for several increasing values of n (and $p = \lfloor \gamma n \rfloor$).

1. Plot the empirical histogram of the eigenvalues of $\hat{\Sigma}_n$.
2. Overlay the theoretical Marchenko–Pastur density for the same γ .
3. Comment on how the agreement between the histogram and the MP density improves as n grows.